

Q1 (20 July 2021 Shift 2)

If time (t), velocity (v), and angular momentum (l) are taken as the fundamental units. Then the dimension of mass (m) in terms of t , v and l is :

(1) $[t^{-1}v^1l^{-2}]$

(2) $[t^1v^2l^{-1}]$

(3) $[t^{-2}v^{-1}l^1]$

(4) $[t^{-1}v^{-2}l^1]$

Q2 (25 July 2021 Shift 2)

The force is given in terms of time t and displacement x by the equation

$$F = A \cos Bx + C \sin Dt$$

The dimensional formula of $\frac{AD}{B}$ is :

(1) $[M^0 L T^{-1}]$

(2) $[ML^2 T^{-3}]$

(3) $[M^1 L^1 T^{-2}]$

(4) $[M^2 L^2 T^{-3}]$

Q3 (27 July 2021 Shift 2)

Match List I with List II.

List-I List-II

(a) Capacitance, C

(i) $M^1 L^1 T^{-3} A^{-1}$

(b) Permittivity of free space, ϵ_0

(ii) $M^{-1} L^{-3} T^4 A^2$

(c) Permeability of free space, μ_0

(iii) $M^{-1} L^{-2} T^4 A^2$

Questions with Answer Keys

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(d) Electric field, E

(iv) $M^1 L^1 T^{-2} A^{-2}$

Choose the correct answer from the options given

below

(1) (a) $\rightarrow$ (iii), (b) $\rightarrow$ (ii), (c) $\rightarrow$ (iv), (d) $\rightarrow$ (i)(2) (a) $\rightarrow$ (iii), (b) $\rightarrow$ (iv), (c) $\rightarrow$ (ii), (d) $\rightarrow$ (i)(3) (a) $\rightarrow$ (iv), (b) $\rightarrow$ (ii), (c) $\rightarrow$ (iii), (d) $\rightarrow$ (i)(4) (a) $\rightarrow$ (iv), (b) $\rightarrow$ (iii), (c) $\rightarrow$ (ii), (d) $\rightarrow$ (i)

Answer Key

Q1 (4)

Q2 (2)

Q3 (1)

Hints and Solutions

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Q1

$$m \propto t^a v^b \ell^c$$

$$m \propto [T]^a [LT^{-1}]^b [ML^2 T^{-1}]^c$$

$$M^1 L^0 T^0 = M^c L^{b+2c} T^{a-b-c}$$

$$\text{comparing powers } c = 1, b = -2, a = -1$$

$$m \propto t^{-1} v^{-2} \ell^1$$

Q2

$$[A] = [MLT^{-2}]$$

$$[B] = [L^{-1}]$$

$$[D] = [T^{-1}]$$

$$\left[\frac{AD}{B}\right] = [ML^2 T^{-3}]$$

Q3

$$q = CV$$

$$[C] = \left[\frac{q}{V}\right] = \frac{(A \times T)^2}{ML^2 T^{-2}}$$

$$= M^{-1} L^{-2} T^4 A^2$$

$$[E] = \left[\frac{F}{q}\right] = \frac{MLT^{-2}}{AT}$$

$$= MLT^{-3} A^{-1}$$

$$F = \frac{q_1 q_2}{4\pi\epsilon_0 r^2}$$

$$[\epsilon_0] = M^{-1} L^{-3} T^4 A^2$$

$$\text{Speed of light } c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$$

$$\mu_0 = \frac{1}{\epsilon_0 c^2}$$

$$[\mu_0] = \frac{1}{[M^{-1} L^{-3} T^4 A^2] [LT^{-1}]^2}$$

$$= [M^1 L^1 T^{-2} A^{-2}]$$